Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Computer Engineering
(NBA Accredited)

Subject- Object Oriented Programming (CO212)
Unit 2 – Overloading and Inheritance
Topic – 2.2 Data Conversion and Type Casting

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Data Conversion & Type Casting

- Type Casting is used to convert the type of a variable, function, object, expression or return value to another type.
- It is the process of converting one type into another.
- In other words converting an expression of a given type into another is called type casting.
- There are two ways of achieving Type Casting

1. Implicit Type Casting (Conversion)
2. Explicit Type Casting (Conversion)

Implicit Type Conversion

- Implicit type conversion is not done by using any operator.
- In other words, the value that gets automatically converted to the specific type to which it is assigned, is known as implicit type conversion.

Explicit Type Conversion

- Explicit conversion can be done using type cast operator.
- In other words, the value that gets converted to the specific type by using type cast operator is known as explicit type conversion.
- In C++, type casting can be done in either of the two ways mentioned below:
- „C"-style casting
- „C++"-style casting

C-style Casting

- In „C" Language, the type casting is made by putting the bracketed name of the required type just before the value of other data type.
- This converts the given data type into bracketed data type.

Example to use ‘C’-style type casting

```
#include<iostream>
int main()
{
    float a;
    int x, y;
    cout<<“\n Enter the Values of x and y:”;
    cin>>x>>y;
    a = (float) x/y;
    cout<<“The Value of a is….”<<a;
    return 0;
}
```

Output:

```
Enter the Values of x and y: 6   4
The Value of a    is…. 1.500000
```

‘C++’-style Type Casting

- C++ permits explicit type conversion of variables or expressions using the type cast operators.
- C++ has the following type cast operators:

1. const_cast
2. static_cast
3. dynamic_cast
4. reinterpret_cast

The const_cast Operator

- A *const_cast operator* is used to add or remove a const to or from a type.

  In general, it is dangerous to use the const_cast operator, because it allows a program to modify a variable that was declared const, but it was not supposed to be modifiable.

The static_cast Operator

- The static-cast operator is used to convert a given expression to the specified type.
- static_cast operator can be used to convert an int to a char.

The dynamic_cast Operator:

- The dynamic_cast operator performs type conversions at run time.

The reinterpret_cast Operator:

- The reinterpret_cast operator changes one data type into another.
- It should be used to cast between incompatible pointer types.
- The reinterpret_cast operator can be used for conversions such as char* to int*, or One_class* to Unrelated_class*.

Thank You..